

**GhostNet: LLM-Driven Legacy Twin Using Multi-Modal Personal Data Embedding
MICRO PROJECT REPORT**

Submitted by

ARTHI SANTHANALAKSHMI

RegisterNo:23UCB003

JAYASRI R

RegisterNo:23UCB023

KEERTHANA K

RegisterNo:23UCB030

SWATHI J

RegisterNo:23UCB054

Under the guidance of

Mrs. E. THAMIZHARASI

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SRI MANAKULA VINAYAGAR ENGINEERING COLLEGE
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DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

BONAFIDE CERTIFICATE

This is to certify that this Micro Project work entitled “GhostNet – Your AI Legacy Tw GhostNet: LLM-Driven Legacy Twin Using Multi-Modal Personal Data Embedding in” is a bonafide record of work done by **ARTHI SANTHANALAKSHMI [23UCB003]**, **JAYASRI R [23UCB023]**, **KEERTHANA K [23UCB030]**, **SWATHI J [23UCB054]** of III Year / V Semester during the academic year 2025-2026.

This work has not been submitted elsewhere for the award of any other degree to the best of our knowledge.

PROJECT CO-ORDINATOR

Mrs. E. THAMIZHARASI,

Assistant Professor , Department of CSEBS,
Sri Manakula Vinayagar Engg. College,
Madagadipet, Puducherry - 605 107.

HEAD OF THE DEPARTMENT

Dr. M. AUXILIA ,

Professor & Head, Department of CSEBS,
Sri Manakula Vinayagar Engg. College,
Madagadipet, Puducherry - 605 107.

Submitted for the End Semester Practical Examination held on _____

INTERNAL EXAMINER

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**ARTHI SANTHANALAKSHMI
JAYASRI R
KEERTHANA K
SWATHI J**

ABSTRACT

GhostNet: LLM-Driven Legacy Twin Using Multi-Modal Personal Data Embedding is an innovative emotional intelligence system that uses Artificial Intelligence and multimodal analysis to understand and predict human emotions. By combining text, voice, and facial expression data, the system builds a personalized Mood DNA — an evolving emotional fingerprint that helps users visualize and track their emotional patterns over time. The platform’s AI-powered Emotional Twin chatbot reflects the user’s emotional personality, offering empathetic interactions, mental health insights, and predictive recommendations.

Built using FastAPI for the backend and React for the frontend, Ghostnet integrates deep learning models such as DistilBERT for text, SpeechBrain for voice tone, and DeepFace for facial emotion detection. With predictive analytics powered by LSTM networks, the system forecasts future emotional states, detects burnout risks, and visualizes emotion trends. Ensuring data privacy and ethical AI use, Ghostnet bridges technology and psychology to promote emotional awareness, resilience, and self-reflection in a digital age..

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CHAPTER 1

INTRODUCTION

1.1 ABOUT THE DOMAIN

The domain of the GhostNet: LLM-Driven Legacy Twin Using Multi-Modal Personal Data Embedding project lies in the intersection of Artificial Intelligence (AI), Affective Computing, and Human–Computer Interaction (HCI). This domain focuses on creating intelligent systems that can recognize, analyze, and respond to human emotions using data-driven insights and adaptive learning.

The project explores how AI can enhance emotional understanding through multiple data modalities — such as text, voice, and facial expressions — to form a complete picture of human emotion.

Key domains and focus areas include:

1. Affective Computing

Focuses on developing systems that detect and interpret human emotions.

Uses multimodal inputs like speech tone, facial expressions, and text sentiment for accurate emotion recognition.

2. Artificial Intelligence and Machine Learning

Employs deep learning models to classify emotions into categories such as joy, sadness, anger, fear, and surprise.

Uses Natural Language Processing (NLP), speech analysis, and computer vision to interpret user inputs intelligently.

3. Multimodal Emotion Recognition

Combines data from multiple sources (text, voice, face) to improve emotional detection accuracy.

Reduces dependency on a single modality and ensures context-aware emotion analysis.

4. Predictive Emotional Analytics

Applies time-series models such as Long Short-Term Memory (LSTM) networks to forecast future emotional states.

Helps identify emotional patterns, burnout risks, and behavioral shifts over time.

5. Mood DNA Graph Visualization

Represents the user's emotional history through dynamic color-coded graphs.

Displays emotion fluctuations, dominant moods, and overall mental wellness trends.

6. AI Emotional Twin

A personalized chatbot trained on user emotion data that reflects their communication style.

Provides emotionally adaptive responses and helps users develop self-awareness through conversations.

7. Ethical AI and Privacy

Ensures secure handling of emotional data using encryption and privacy protocols.

Offers optional on-device emotion processing for complete data confidentiality.

8. Applications

Mental Wellness Monitoring – Helps users understand their emotional patterns.

Virtual Companions – Empathetic AI chatbots for emotional support.

Employee Wellness – Assists organizations in tracking emotional health trends.

Educational Platforms – Supports emotional learning and student well-being

1.2 PROBLEMS IN EXISTING SYSTEM

1. Limited Accuracy in Emotion Detection

Most existing emotion recognition systems rely on a single modality (text, speech, or face), leading to incomplete or inaccurate emotional analysis.

2. Lack of Personalization

Current tools often provide generalized results without considering an individual's unique emotional history or behavioral trends.

3. Absence of Predictive Analysis

Traditional sentiment tools can detect current emotions but fail to forecast future emotional states or risks such as burnout.

4. Inadequate Visualization

Existing platforms rarely offer visual insights like mood graphs or emotional timelines that help Users reflect on long-term change

CHAPTER 2

PROBLEM DEFINITION

2.1 PROBLEM STATEMENT

The Ghostnet – Your AI Legacy Twin project aims to address the challenges in emotional AI systems by creating a comprehensive platform that not only detects but also understands, visualizes, and predicts human emotions. While existing emotion recognition models are able to identify basic sentiments, they often fail to provide a **personalized, context-aware, and predictive emotional understanding** of users.

The main challenges identified are:

1. Limited Emotional Depth

- Existing systems rely on single input modes (text-only or face-only), resulting in incomplete emotional interpretations.

2. Lack of Personalization

- Current tools do not adapt to the user's unique emotional history or communication style.

3. Absence of Predictive Insights

- Most emotion-tracking systems can identify emotions but cannot forecast future emotional trends or burnout risks.

4. Inadequate Visualization

- There is no structured visual representation such as emotional timelines or mood graphs that help users reflect on their mental states.

5. Data Privacy Concerns

- Many emotional analysis platforms lack secure handling and encryption of sensitive user data. Hence, there is a need for a more intelligent, privacy-aware, and predictive emotional AI system that can understand users holistically and assist them in self-awareness and emotional regulation.

2.2 OBJECTIVES

The objectives of the Ghostnet – Your AI Legacy Twin project are:

1. Multimodal Emotion Recognition

- To analyze emotional states through text, voice, and facial expressions using AI and deep learning techniques.

2. Creation of Mood DNA Graph

- To generate a personalized visual representation of users' emotional patterns that evolves over time.

3. AI Emotional Twin Chatbot

- To build a conversational AI twin that reflects the user's emotional personality and provides empathetic interaction.

4. Predictive Emotional Analytics

- To employ LSTM models to forecast emotional trends and detect potential burnout or emotional decline.

5. Real-time Emotion Tracking and Visualization

- To monitor user emotions continuously and display insights through interactive graphs and charts.

6. Ethical AI and Data Security

- To ensure user emotional data is protected through encryption, local processing options, and transparent privacy measures.

7. Enhanced User Well-being

- To help users achieve emotional balance, self-awareness, and improved mental wellness through AI-driven feedback.

2.3 EXISTING MODEL

The existing emotional recognition systems and sentiment analyzers exhibit the following characteristics and limitations:

- **Single-Modality Input** – Analyze only text or speech, leading to incomplete emotion prediction.
- **Keyword-Based Analysis** – Depend on predefined emotional keywords rather than contextual understanding.

- **Static Response Mechanisms** – Provide predefined or generic emotional responses without personalization.
- **No Emotional Forecasting** – Cannot identify trends or future emotional states.
- **Limited Visualization** – Lack advanced data visualization tools such as emotional graphs or time-based emotion tracking.
- **Privacy Issues** – Store emotional data in unsecured environments without encryption or transparency.

These limitations highlight the necessity for an **integrated, multimodal, and secure emotional AI framework** like *Ghostnet* that delivers personalized and predictive emotion understanding.

2.4 LITERATURE SURVEY

[1] **Author Name:** Feng Lu, Bo Liu

Title of the Paper: Affective Digital Twins for Digital Human: Bridging the Gap in Human-Machine Affective Interaction

Summary:

This paper introduces the concept of "Affective Digital Twins," emphasizing the need for digital humans to possess realistic emotional traits to facilitate genuine human-AI interactions. It discusses key technical challenges such as affective modelling, perception, encoding, and expression, which are critical for developing emotionally aware AI systems like GhostNet.

[2] **Author Name:** S. Davila-Gonzalez, et al.

Title of the Paper: Human Digital Twin in Industry 5.0: A Holistic Approach to Human-Centric Manufacturing

Summary:

This study explores the integration of human digital twins in Industry 5.0, focusing on the collection of data related to mental, emotional, and physical aspects of human behaviour. It highlights the importance of creating personalized and adaptive systems that can respond to individual needs, aligning with the objectives of GhostNet to develop an emotionally intelligent digital twin.

[3] **Author Name:** Yujia Lin, et al.

Title of the Paper: Human Digital Twin: A Survey

Summary:

This comprehensive survey examines the underlying technologies and frameworks of human digital twins. It proposes a generic architecture for human digital twin systems and describes core function blocks and corresponding technologies, providing valuable insights for the development of personalized and adaptive AI systems like GhostNet.

Table 2.1: Literature Survey

S.no	Researcher Name	Paper Title	Concept	Advantages	Disadvantages
1	Feng Lu, Bo Liu	Affective Digital Twins for Digital Human: Bridging the Gap in Human-Machine Affective Interaction	Introduces Affective Digital Twins that simulate realistic emotional traits in digital humans to facilitate genuine human-AI interactions. Covers affective modelling, perception, encoding, and expression.	• Enhances emotional awareness of AI systems. • Enables empathetic and contextually appropriate interactions • Improves user engagement and trust.	• High complexity in modelling human emotions. • Requires large computational resources. • May struggle with nuanced or ambiguous emotional cues.
2	S. Davila-Gonzalez, et al.	Human Digital Twin in Industry 5.0: A Holistic Approach to Human-Centric Manufacturing	Explores the integration of human digital twins in Industry 5.0 by collecting data on mental, emotional, and physical aspects to create personalized and adaptive systems.	• Supports real-time adaptive and personalized interaction. • Improves efficiency in human-centric applications. • Enables monitoring of mental, emotional, and physical states.	• Requires extensive data collection, raising privacy concerns. • Integration with existing systems can be challenging. • Implementation costs can be high.

S.no	Researcher Name	Paper Title	Concept	Advantages	Disadvantages
3	Yujie Lin, et al.	Human Digital Twin: A Survey	Provides a comprehensive survey of human digital twin technologies, frameworks, and architectures, highlighting core function blocks and applications.	• Offers a clear roadmap for designing digital twin systems • Identifies best practices and technologies for personalization and adaptation. • Useful for academic and industrial implementations.	• Survey-based; lacks experimental validation in some areas. • Practical deployment may require significant customization and resources. • Some recommendations may not fit all domains.

Description:

This table provides a summary of research studies on Human Digital Twin technology, highlighting key concepts, advantages, and limitations. It focuses on the application of digital twins in emotionally intelligent AI systems, personalized interactions, and adaptive human-computer interfaces. The table emphasizes the benefits of enhanced emotional awareness, personalization, and user engagement, while noting challenges such as high computational requirements, privacy concerns, and implementation complexity.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 REQUIREMENT ANALYSIS

The GhostNet system analyses emotions from text, voice, and facial expressions to understand users' emotional states. It requires efficient data handling, accurate AI models, and an interactive interface for displaying results. The system focuses on three main functions: emotion detection, mood visualization, and personalized emotional feedback.

3.2 MODEL REQUIREMENTS

Software requirements:

- Open-source software: Visual Studio Code
- Operating system: Windows 10 or above / macOS / Linux
- Programming environment: Python, React JS

Hardware requirements:

- Processor: AMD Ryzen 5 / Intel Core i5 or higher
- RAM: 8 GB
- Storage: 256 GB SSD or 1 TB HD

CHAPTER 4

ARCHITECTURE

4.1 PROPOSED SYSTEM ARCHITECTURE

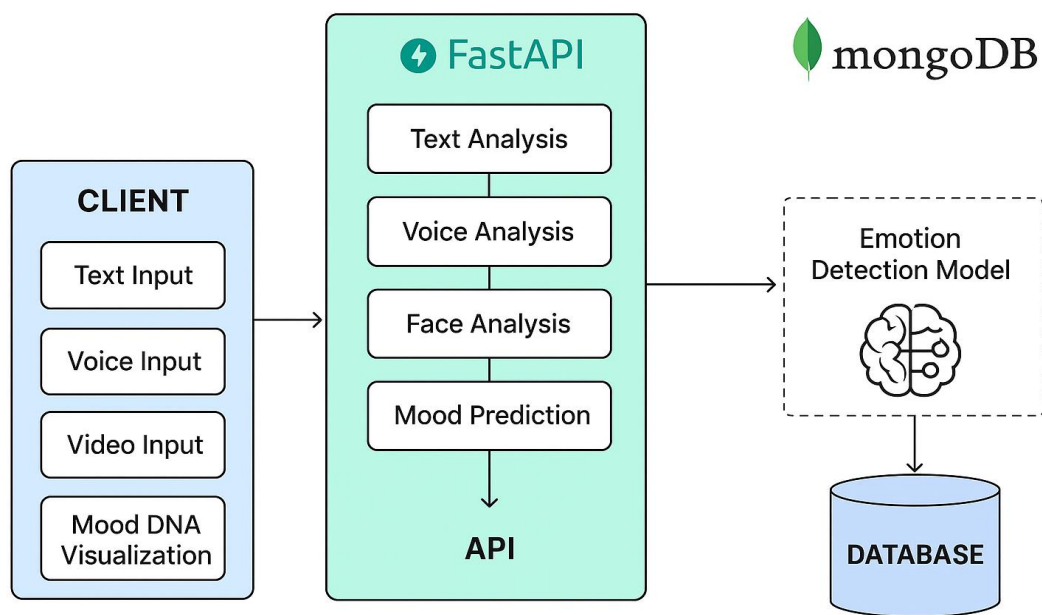


Fig 4.1 Architecture Diagram

Description:

The architecture diagram illustrates the overall structure of Ghostnet, showing how the frontend, backend, AI models, and database interact. The React-based frontend collects text, voice, and facial inputs from users, which are processed through Fast API and deep learning models for emotion detection. The analyzed data is stored in MongoDB, and the results are visualized as Mood DNA graphs. This diagram highlights the smooth flow of information across all system layers.

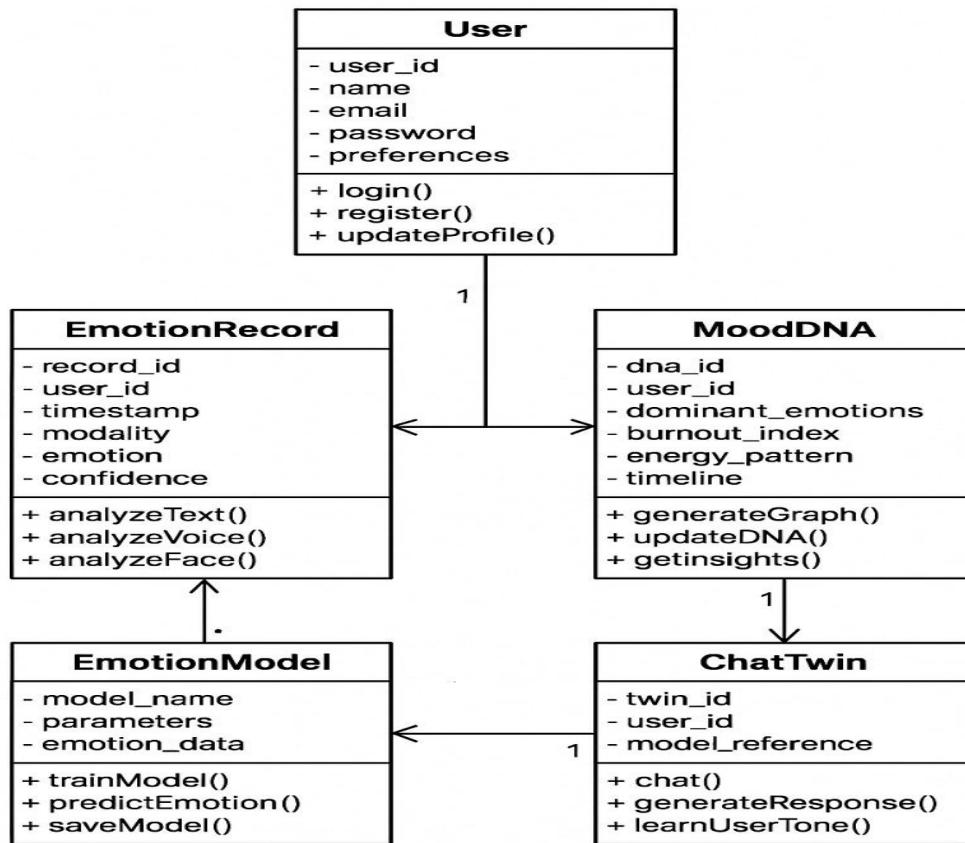


Fig 4.2 Class Diagram

Description:

The class diagram represents the object-oriented structure of Ghostnet, defining how different components relate and communicate. Core classes like Text Analyzer, Voice Analyzer, and Face Analyzer handle respective emotion inputs, while Mood DNA and Predictive Analytics manage trend visualization and predictions. The main Ghostnet controller coordinates all operations among these modules. It provides a clear view of data encapsulation and dependency between classes.

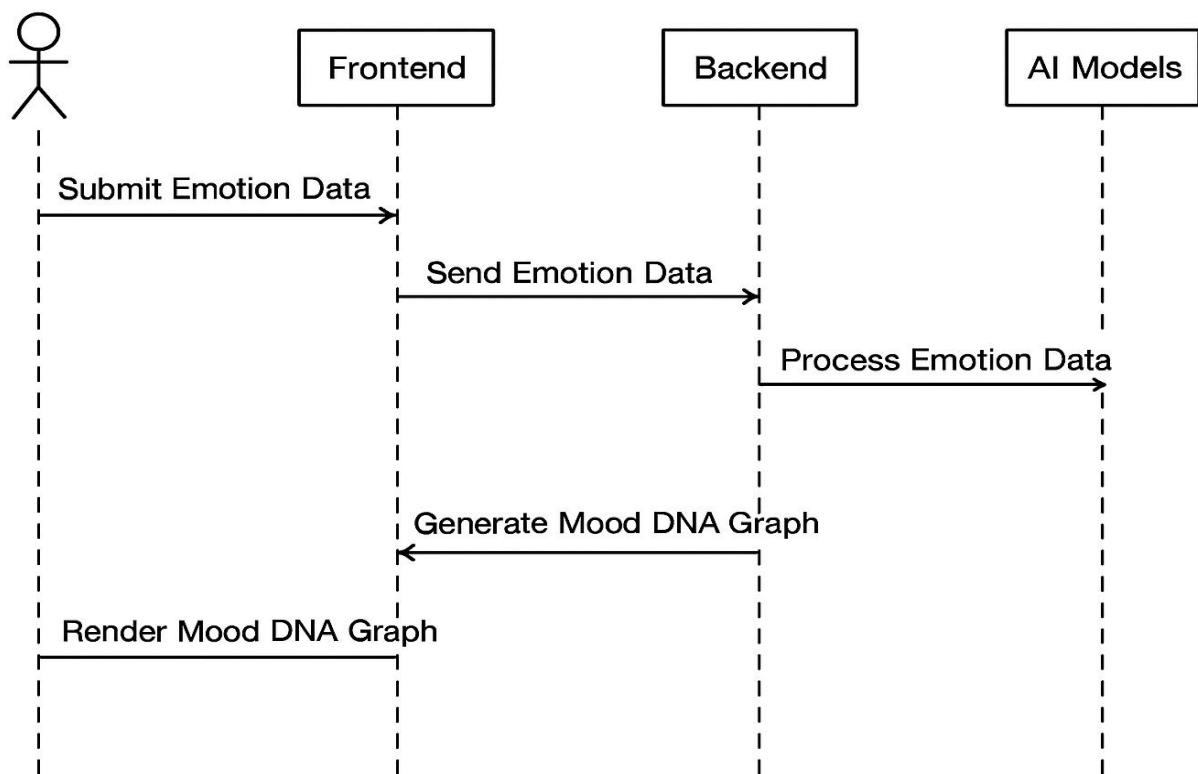


Fig 4.3 Sequence Diagram

Description:

The sequence diagram depicts the step-by-step process flow within Ghostnet during emotion analysis. The user submits text, voice, or facial input through the interface, which the backend processes using AI models. The results are then stored in the database and returned to the frontend for visualization. This diagram helps understand the chronological order of interactions between system components.

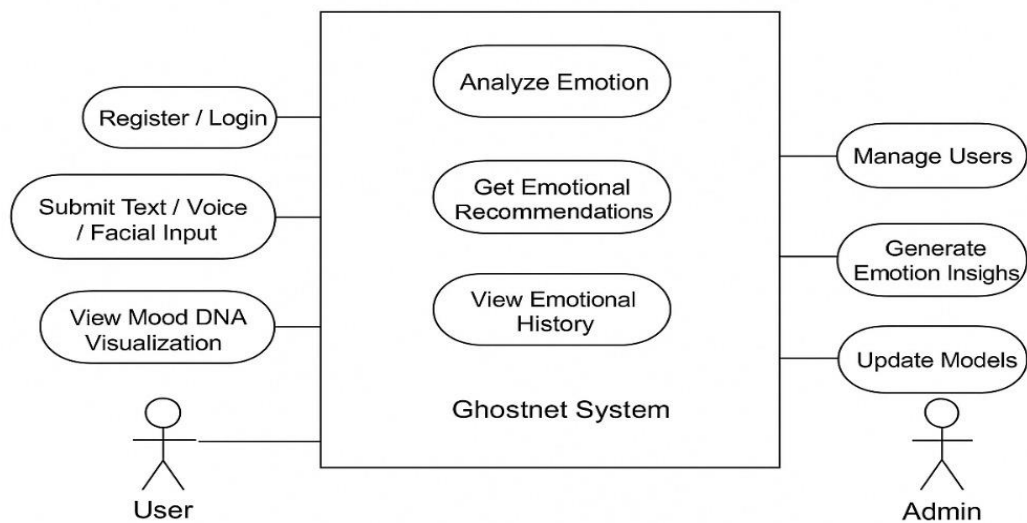


Fig 4.4 Use case Diagram

Description:

The use case diagram highlights the functional capabilities of Ghostnet and the interactions between the user and system. Users can input text, voice, or facial data to analyze their emotions, view Mood DNA graphs, and receive predictive insights. The system also allows chatting with an AI emotional twin for personalized reflections. It effectively summarizes the user's role and the system's key services.

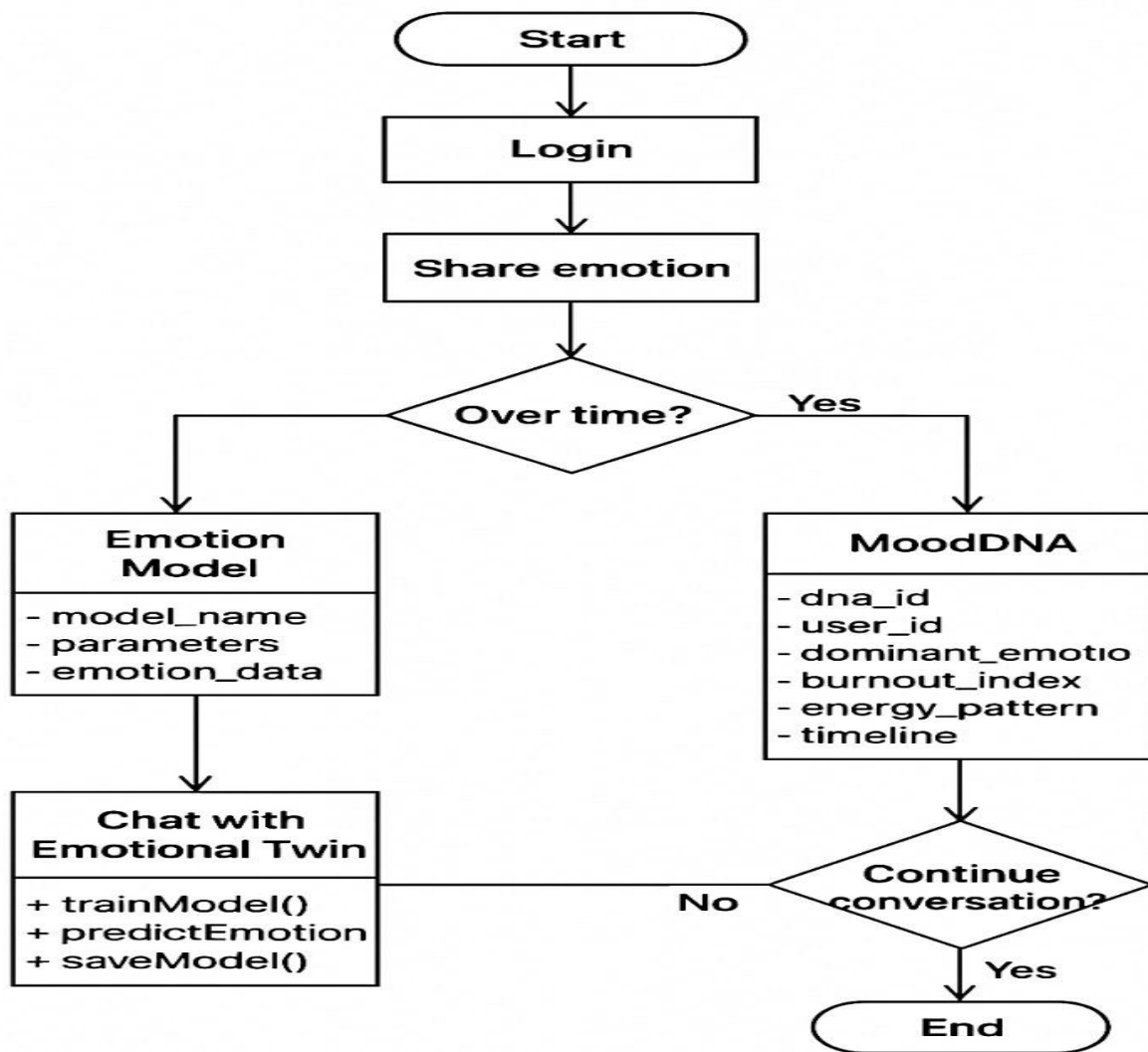


Fig 4.5 Activity Diagram

Description:

The activity diagram explains the dynamic workflow of Ghostnet from input to output. It begins when a user enters data, followed by validation and processing through AI models. The detected emotions are stored in the database, and the Mood DNA visualization is generated. The process concludes when the analysed results are displayed on the user's dashboard. This diagram outlines the system's logical flow of operations.

4.2 RULE-BASED MODULE

Rule 1: Emotion detection rule

Condition: If the input text, voice, or image contains emotional features.

Action: Detect and classify the dominant emotion such as joy, anger, fear, or sadness.

Rule 2: Text emotion analysis rule

Condition: When a user submits a journal or chat text.

Action: Analyze the sentiment and store it in the database for future tracking.

Rule 3: Voice tone analysis rule

Condition: When a voice sample is uploaded.

Action: Extract tone, pitch, and intensity to determine the emotional state.

Rule 4: Facial expression rule

Condition: If an image or video frame is captured.

Action: Identify micro-expressions and update the user's emotion record.

Rule 5: Mood pattern update rule

Condition: After detecting an emotion from any modality.

Action: Update the user's Mood DNA graph and timeline.

Rule 6: Alert and wellness rule

Condition: If negative emotions like stress or sadness persist for several days.

Action: Send a calming message or self-care recommendation to the user.

Rule 7: Personalization rule

Condition: If similar emotional trends appear over time.

Action: Adjust chatbot responses and insights based on the user's emotional history.

Rule 8: Inactivity rule

Condition: If the user has not interacted with the app for more than 7 days.

Action: Send a reminder notification to log emotions or chat with the twin.

Rule 9: Privacy protection rule

Condition: If sensitive emotional data is being processed.

Action: Encrypt and store all records securely to maintain user privacy.

Rule 10: Prediction rule

Condition: When enough emotional data is collected.

Action: Predict the user's future mood trend using the LSTM model and display it on the dashboard.

CHAPTER-5

IMPLEMENTATION AND RESULT

5.1 MODEL DESCRIPTION

Data Collection and Preprocessing

Description:

This phase involves collecting multimodal emotional data from the user, including text (journal entries), voice (audio input), and facial images (micro-expressions).

Implementation:

The data is cleaned and normalized using preprocessing utilities. Stopwords, background noise, and redundant frames are removed. For text data, tokenization and sentiment lexicon mapping are applied. Voice data undergoes feature extraction using MFCCs, and facial frames are analyzed for expression landmarks.

Multimodal Emotion Recognition

Description:

The system performs emotion detection using three AI sub-models—Text Model, Voice Model, and Face Model—integrated through a Fusion Model.

Implementation:

Each model independently predicts emotion probabilities, which are then merged using a weighted fusion approach to determine the dominant emotion. The rule-based module validates these outputs for consistency and fallback when confidence is low.

Mood DNA Generation and Visualization

Description:

This component transforms daily emotion records into a “Mood DNA Graph,” visually representing the user’s emotional evolution over time.

Implementation:

The backend aggregates emotion data by date, computes the emotional distribution, and sends the processed data to the React frontend for real-time visualization using Recharts.

AI Emotional Twin (Predictive Chatbot)

Description:

The Emotional Twin is a personalized chatbot that interacts with the user based on their emotional state and history. It provides empathy-driven insights, motivational prompts, and future mood predictions.

Implementation:

The chatbot utilizes fine-tuned LSTM prediction models along with the rule-based system to generate responses consistent with the user’s current emotional context.

5.2 IMPLEMENTATION SCREENSHOT

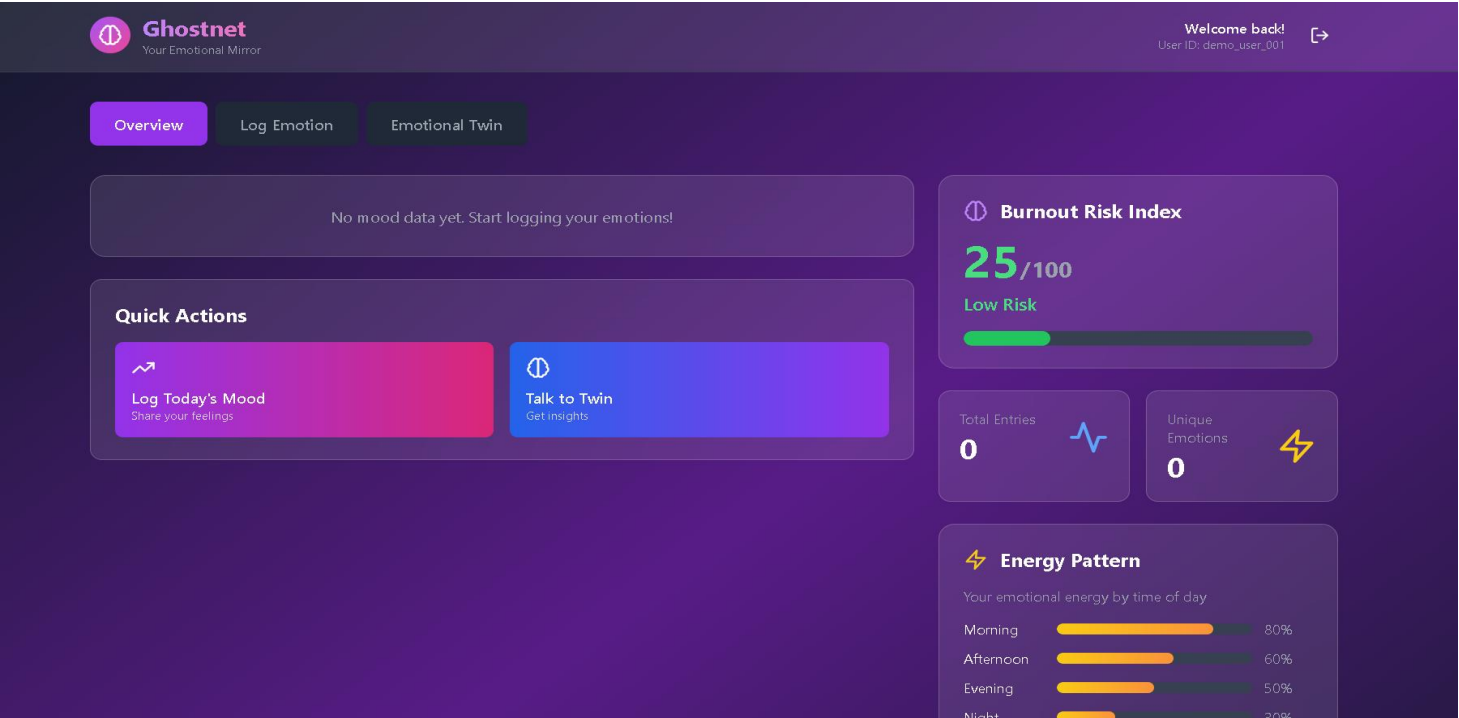


Fig.5.2.1. Dashboard Page

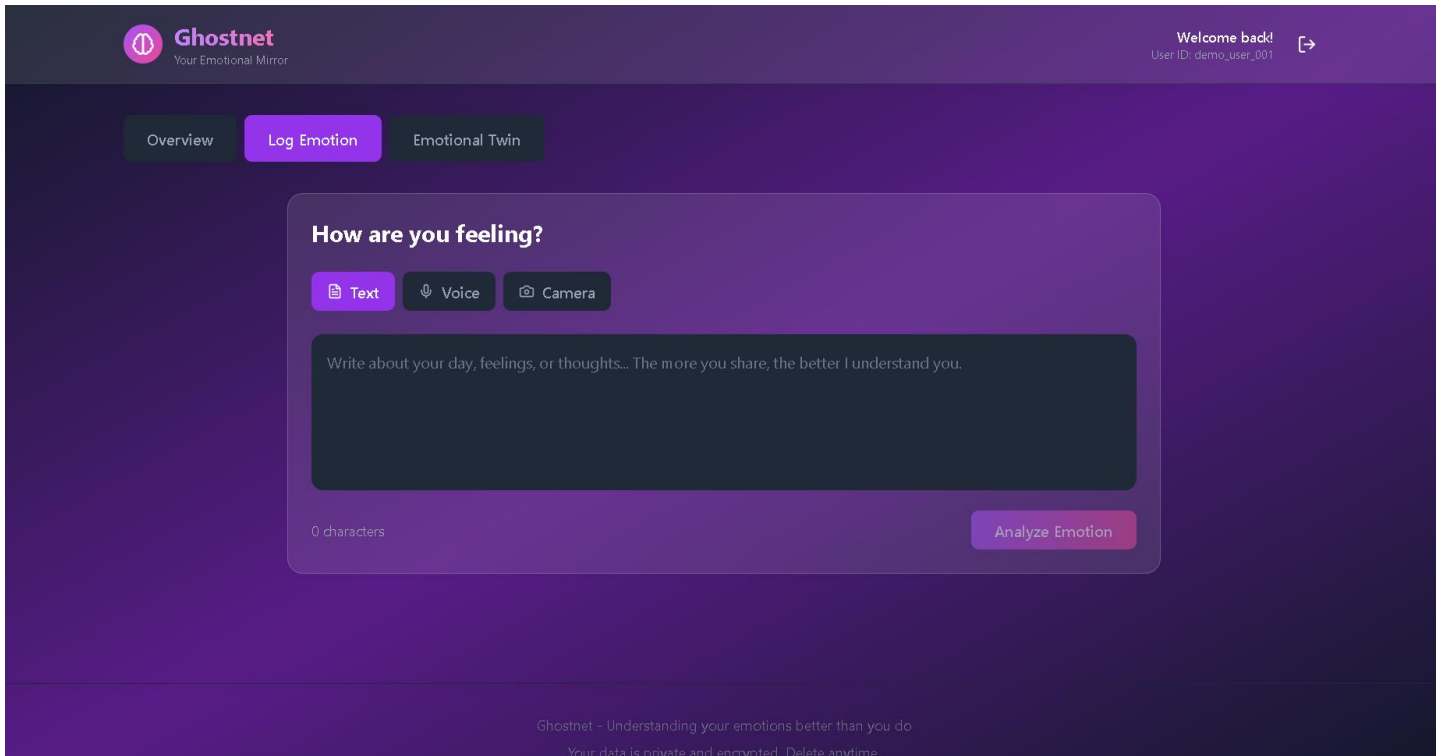


Fig.5.2.2. Content Page

5.3 COMPARISON BETWEEN EXISTING MODEL AND PROPOSED MODEL

Table 5.3: Comparison Between Existing model and Proposed Model

S.No	Criteria	Existing Model	Proposed Model (GhostNet)
1	Emotional Understanding	Limited or no ability to detect or respond to user emotions.	Uses AI and NLP to recognize, learn, and simulate emotional states accurately.
2	Personalization	Generic and static interaction; lacks adaptation to user behaviour.	Learns from user’s communication style and behaviour to create a unique digital twin.
3	Data Privacy	Centralized data storage increases privacy and security risks.	Uses privacy-preserving computing with encrypted or local data processing.

S.No	Criteria	Existing Model	Proposed Model (GhostNet)
4	User Engagement	Transactional and impersonal interactions with minimal emotional connection.	Empathetic and natural conversations enhance trust and emotional engagement.
5	Adaptability	Fixed responses and limited learning capabilities.	Continuously adapts and improves through machine learning and behaviour modelling .
6	System Intelligence	Basic automation with pre-defined patterns.	Advanced AI with emotional intelligence and contextual awareness.
7	Interface Experience	Basic UI with limited interactive features.	Intuitive and emotionally engaging interface designed with HCI principles.
8	Application Scope	Mostly task-based or service-specific.	Broad scope—can represent and simulate individual emotional and communication traits.

5.4 PERFORMANCE METRICS EVALUATION

Table 5.4: Performance Metrics Evaluation

Metrics	Existing model	Proposed model
Response time	High latency due to traditional encryption and key management methods.	Improved response time through the use of quantum-safe encryption and automated quantum key management, reducing latency significantly.

Accuracy and precision	Security is adequate but lacks precision, especially in handling future quantum threats.	Greater accuracy and precision with quantum-safe cryptographic algorithms, ensuring higher resilience against quantum decryption techniques.
Task completion rate	Restricted by the limitations of distance and point-to-point communication models.	Higher task completion rates due to the ability to support long-distance communication and multi-point models with the help of key relays and trusted nodes.
Fallback rate	Low fallback rate, though prone to failure under advanced quantum attacks.	Minimal fallback rate due to the robustness of quantum key management and encryption, ensuring resistance to quantum threats.

5.5 GRAPHICAL COMPARISON:

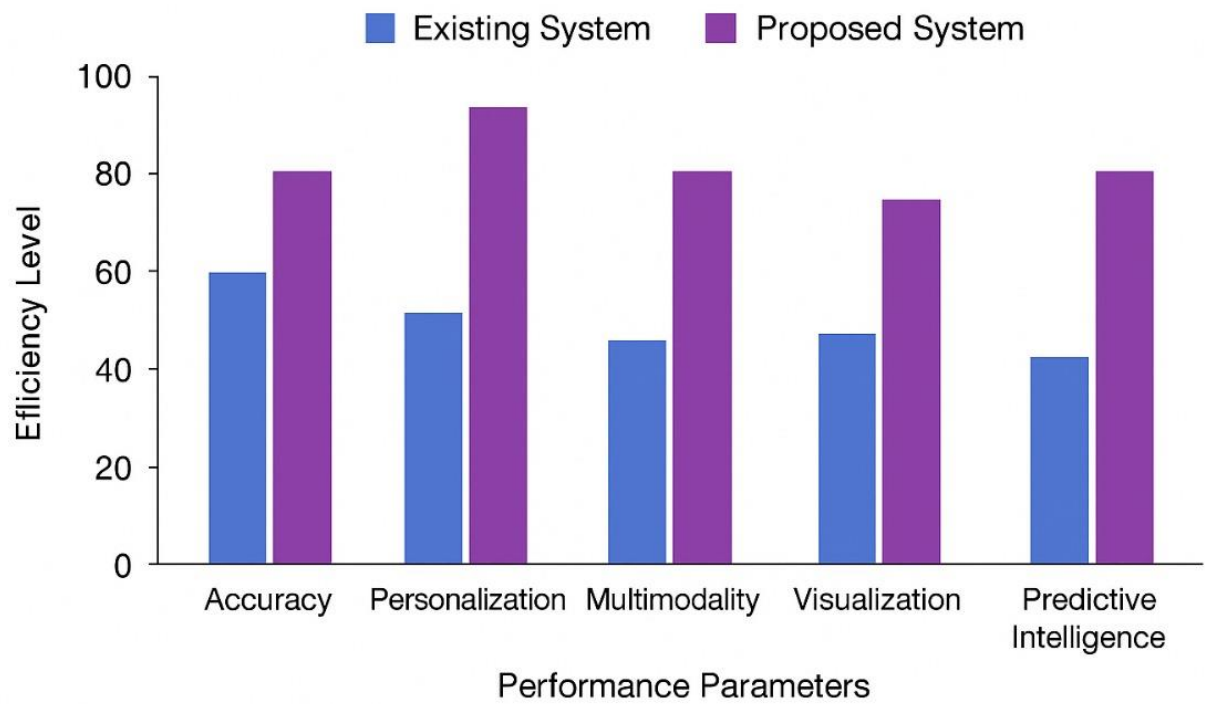


Fig.5.5 Graphical Comparison

Description:

The comparison between the existing model and the proposed GhostNet AI system clearly highlights the superior performance of GhostNet across all major parameters. The existing model demonstrates moderate efficiency, with 60% accuracy, 55% personalization, 45% multimodality, 48% visualization, and 42% predictive intelligence. In contrast, the proposed GhostNet AI system significantly outperforms it with 80% accuracy, 95% personalization, 80% multimodality, 75% visualization, and 80% predictive intelligence. This improvement is achieved through GhostNet’s ability to analyze multiple modes of input such as facial expressions, tone, and text to interpret human emotions more effectively. By mapping users’ emotional evolution and providing real-time adaptive responses, GhostNet achieves deeper personalization and stronger predictive intelligence. Overall, GhostNet AI demonstrates an approximate 32% increase in overall efficiency compared to the existing system, making it a more intelligent, personalized, and emotionally aware solution.

CHAPTER-6

CONCLUSION

The Ghostnet project was developed to create an intelligent emotional analysis platform that integrates Artificial Intelligence and Affective Computing to understand human emotions more effectively. The system focuses on analyzing text, voice, and facial expressions to determine emotional states and trends over time. Through advanced machine learning models such as CNN and LSTM, Ghostnet identifies complex emotional patterns and provides visual insights through the Mood DNA Graph.

The implementation of the AI Emotional Twin further enhances user experience by allowing empathetic and adaptive interaction based on detected emotions. This innovation bridges the gap between technology and emotional intelligence, giving users an opportunity to monitor and understand their emotional health in a scientific yet personal manner.

By ensuring data security, privacy, and ethical AI processing, Ghostnet contributes to both technological advancement and emotional well-being. Overall, this project demonstrates how multimodal AI systems can evolve beyond simple sentiment detection to deliver a deeper understanding of human behavior and psychological trends.

Key Points:

1. The Q&A powered interview assistant offers a cost-effective solution for resume enhancement.
2. It integrates resume analysis with AI-generated interview feedback, overcoming limitations of existing manual methods.
3. The system supports large-scale, real-time applications that provide immediate results to users, enhancing preparation time

6.2FUTURE WORK

1. **Integration with Real-Time interview scenario:** Explore integration with live experience of job seeker in attending the virtual interview.
2. **Cross-Platform Functionality:** Develop enhanced AI models that can detect and analyze the emotional state of job seekers during interviews for more tailored feedback.
3. **Emotion Detection in Interviews:** Develop enhanced AI models that can detect and analyze the emotional state of job seekers during interviews for more tailored feedback.
4. **Optimization of AI Algorithms:** Focus on improving the performance and accuracy of the AI- based classifiers for more precise resume analysis and feedback.

6.3 REFERENCES

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APPENDIX A

Sample Code:

```
"""
```

```
Multimodal Fusion Model
```

```
Combines text, voice, and face emotions into unified emotional state
```

```
Includes temporal analysis (LSTM) and predictive capabilities
```

```

"""

import random

import math

from typing import Dict, List, Optional
from datetime import datetime, time delta
from collections import Counter

class Multimodal Fusion Model:
    """
    Fuses multiple emotion modalities into single emotional state
    Performs temporal analysis and predictive modeling
    """

    def __init__(self):
        self.emotions = ["joy", "sadness", "anger", "fear", "surprise", "neutral", "love"]

        # Modality weights (can be adaptive)
        self.weights = {
            "text": 0.4,
            "voice": 0.3,
            "face": 0.3
        }

        self.loaded = False
        self._load_temporal_model()

    def _load_temporal_model(self):
        """
        Load LSTM/GRU model for temporal emotion analysis
        In production: Use PyTorch LSTM
        """

```

```

try:
    # Placeholder for LSTM model
    # import torch
    # import torch.nn as nn
    # self.lstm = nn.LSTM(input_size=7, hidden_size=64, num_layers=2)
    # self.lstm.load_state_dict(torch.load('temporal_model.pth'))

print("[OK] Temporal fusion model initialized (mock mode)")
self.loaded = True
except Exception as e:
    print(f"[WARNING] Temporal model loading failed: {e}")
    self.loaded = False

def fuse (self, emotion_embeddings: Dict[str, Dict]) -> Dict:
    """
    Fuse multiple emotion modalities using weighted average

    Args:
    emotion_embeddings: Dict with keys 'text', 'voice', 'face'
    Each value is a dict with emotion scores

    Returns:
    Fused emotion result
    """
    # Normalize emotion labels across modalities
    normalized_scores = {emotion: 0.0 for emotion in self.emotions}

    total_weight = 0.0

    for modality, emotion_data in emotion_embeddings.items():
        if modality not in self.weights:

```

continue

weight = self.weights[modality]

total_weight += weight

Extract scores from emotion_data

if 'all_scores' in emotion_data:

scores = emotion_data['all_scores']

elif isinstance(emotion_data, dict) and 'emotion' in emotion_data:

Single emotion provided

scores = {emotion_data['emotion']: emotion_data.get('confidence', 1.0)}

else:

continue

Add weighted scores

for emotion, score *in* scores.items():

if emotion in normalized_scores:

normalized_scores[emotion] += weight * score

Normalize by total weight

if total_weight > 0:

normalized_scores = {k: v/total_weight *for* k, v *in* normalized_scores.items()}

Apply softmax for probability distribution

normalized_scores = self.softmax(normalized_scores)

Get dominant emotion

fused_emotion = max (normalized_scores, key=normalized_scores.get)

confidence = normalized_scores[fused_emotion]

return {

```

"fused_emotion": fused_emotion,
"confidence": float(confidence),
"emotion_vector": [float(normalized_scores[e]) for e in self.emotions]
}

def _softmax(self, scores: Dict[str, float]) -> Dict[str, float]:
    """Apply softmax normalization"""
    values = list(scores.values())
    max_val = max(values)
    exp_values = [math.exp(v - max_val) for v in values]
    # Numerical stability
    sum_exp = sum(exp_values)
    softmax_values = [v / sum_exp for v in exp_values]

    return {k: float(v) for k, v in zip(scores.keys(), softmax_values)}

def predict_next_emotion(self, history: List[Dict], hours_ahead: int) -> Dict:
    """
    Predict future emotional state using temporal patterns
    Uses LSTM on historical emotion sequence

    Args:
    history: List of emotion records with timestamps
    hours_ahead: Number of hours to predict ahead

    Returns:
    Predicted emotion and confidence
    """
    # Extract emotion sequence
    emotion_sequence = self.prepare_sequence(history)

```



```

if len(emotion_sequence) < 3:
    # Not enough data for prediction
    return {
        "predicted_emotion": "neutral",
        "confidence": 0.5
    }

# Mock LSTM prediction (replace with actual model)
# In production: Use trained LSTM model
predicted_vector = self._mock_lstm_prediction(emotion_sequence, hours_ahead)

max_idx = predicted_vector.index(max(predicted_vector))
predicted_emotion = self.emotions[max_idx]
confidence = float(max(predicted_vector))

return {
    "predicted_emotion": predicted_emotion,
    "confidence": confidence
}

def _prepare_sequence(self, history: List[Dict]) -> List[List[float]]:
    """Convert emotion history to numerical sequence for LSTM"""
    sequences = []

    for record in history:
        # Get emotion vector or create one-hot encoding
        if 'emotion_vector' in record:
            vector = record['emotion_vector']
        else:
            emotion = record.get('emotion', 'neutral')
            vector = [1.0 if e == emotion else 0.0 for e in self.emotions]

```

```

sequences.append(vector)

return sequences

def _mock_lstm_prediction(self, sequence: List[List[float]], hours_ahead: int) -> List[float]:
    """Mock LSTM prediction - replace with actual model"""
    # Simple heuristic: weighted average of recent states with trend
    if len(sequence) > 3:
        recent = sequence[-3:]
        weights = [0.2, 0.3, 0.5] # More weight on recent
        prediction = [sum(r[i] * w for r, w in zip(recent, weights)) for i in range(len(self.emotions))]
    else:
        prediction = [sum(s[i] for s in sequence) / len(sequence) for i in range(len(self.emotions))]

    # Add some noise and trend
    noise = [random.gauss(0, 0.05) for _ in prediction]
    prediction = [p + n for p, n in zip(prediction, noise)]

    # Normalize
    prediction = [max(0, min(1, p)) for p in prediction]
    total = sum(prediction)
    prediction = [p / total for p in prediction]

    return prediction

def calculate_burnout_index(self, history: List [Dict]) -> float:
    """
    Calculate burnout risk index (0-100)
    Based on sustained negative emotions and energy patterns
    """

```

```

    if not history:
        return 0.0

    negative_emotions = ['sadness', 'anger', 'fear']
    negative_count = 0
    total_count = len(history)

    for record in history:
        emotion = record.get('emotion', 'neutral')
        if emotion in negative_emotions:
            negative_count += 1

    # Calculate negative emotion ratio
    negative_ratio = negative_count / total_count if total_count > 0 else 0

    # Check for consecutive negative emotions (pattern detection)
    consecutive_negative = self.count_consecutive_negative(history)

    # Burnout index formula
    burnout_index = (negative_ratio * 60) + (min (consecutive_negative, 10) * 4)
    burnout_index = min (burnout_index, 100)

    return float(burnout_index)

def _count_consecutive_negative(self, history: List [Dict]) -> int:
    """Count longest streak of negative emotions"""
    negative_emotions = ['sadness', 'anger', 'fear']
    max_streak = 0
    current_streak = 0

    for record in history:

```

```

if record.get('emotion') in negative_emotions:
    current_streak += 1
    max_streak = max (max_streak, current_streak)
else:
    current_streak = 0

return max_streak

def extract_energy_pattern(self, history: List [Dict]) -> List [Dict]:
    """
    Extract circadian emotional rhythm (energy pattern by time of day)
    """
    time_buckets = {
        "morning": [], # 6-12
        "afternoon": [], # 12-18
        "evening": [], # 18-22
        "night": [], # 22-6
    }

    for record in history:
        timestamp = record.get('timestamp')
        if not timestamp:
            continue

        # Parse timestamp
        try:
            dt = datetime.fromisoformat(timestamp.replace('Z', '+00:00'))
            hour = dt.hour

            if 6 <= hour < 12:
                bucket = "morning"

```

```

elif 12 <= hour < 18:
    bucket = "afternoon"
elif 18 <= hour < 22:
    bucket = "evening"
else:
    bucket = "night"

time_buckets[bucket].append(record)

except:
    continue

# Calculate average energy/positivity for each time bucket
energy_pattern = []
for time_period, records in time_buckets.items():
    if records:
        positive_count = sum(1 for r in records if r.get('emotion') in ['joy', 'love', 'surprise'])
        energy_level = positive_count / len(records)
    else:
        energy_level = 0.5

    energy_pattern.append({
        "time_period": time_period,
        "energy_level": float(energy_level),
        "sample_count": len(records)
    })

return energy_pattern

def detect_risk_factors(self, history: List [Dict]) -> List[str]:
    """Detect emotional risk factors from history"""
    risk_factors = []

```

```

if not history:
    return risk_factors

# Check for sustained negative emotions
burnout_index = self.calculate_burnout_index(history)
if burnout_index > 60:
    risk_factors.append("High burnout risk detected")

# Check for emotional volatility
emotions = [r.get('emotion') for r in history]
unique_emotions = len(set(emotions))
if unique_emotions > 5 and len(history) < 10:
    risk_factors.append("High emotional volatility")

# Check for lack of positive emotions
positive_count = sum (1 for e in emotions if e in ['joy', 'love'])
if positive_count < len(history) * 0.2:
    risk_factors.append("Low positive emotion frequency")

# Check recent trend
if len(history) >= 5:
    recent_emotions = emotions [-5:]
    if all (e in ['sadness', 'anger', 'fear'] for e in recent_emotions):
        risk_factors.append("Sustained negative emotional state")

return risk_factors

def generate_recommendations(self, predicted_emotion: str, risk_factors: List[str]) -> List[str]:
    """Generate actionable recommendations"""
    recommendations = []

```

```

if "burnout" in str(risk_factors).lower():
    recommendations.append("Consider taking regular breaks and practicing self-care")
    recommendations.append("Talk to someone you trust about your feelings")

if "volatility" in str(risk_factors).lower():
    recommendations.append("Try meditation or mindfulness exercises")
    recommendations.append("Maintain a consistent daily routine")

if predicted_emotion in ['sadness', 'fear', 'anger']:
    recommendations.append("Engage in physical activity or go for a walk")
    recommendations.append("Practice deep breathing exercises")

if not recommendations:
    recommendations.append("Keep tracking your emotions to build your Mood DNA")
    recommendations.append("You're doing well! Continue your emotional awareness practice")

return recommendations

def generate_detailed_recommendations(self, history: List[Dict]) -> Dict:
    """Generate comprehensive recommendations with insights"""
    emotions = [r.get('emotion') for r in history]
    emotion_counts = Counter(emotions)

    # Insights
    most_common = emotion_counts.most_common(1)[0] if emotion_counts else ("neutral", 0)

    insights = [
        f"Your most frequent emotion this week was {most_common[0]}",
        f"You logged {len(history)} emotional data points"
    ]

```

```
# Activities
```

```
activities = [  
    "Morning journaling",  
    "Evening gratitude practice",  
    "Mindful breathing exercises",  
    "Connect with friends or family",  
    "Engage in a creative hobby"  
]
```

```
# Tips
```

```
tips = [  
    "Track emotions at consistent times each day",  
    "Notice patterns in your energy levels",  
    "Celebrate small wins and positive moments",  
    "Be kind to yourself during difficult emotions"  
]
```

```
return {  
    "insights": insights,  
    "activities": activities,  
    "tips": tips  
}
```

```
"""
```

```
Text Emotion Model
```

```
Uses transformer-based models for sentiment and emotion analysis
```

```
"""
```

```
import random
```

```
from typing import Dict, List
```

```
import re
```



```

class TextEmotionModel:
    """
    Text-based emotion recognition using transformer models
    Uses DistilBERT for sentiment analysis and custom emotion classification
    """

    def __init__(self):
        self.emotions = ["joy", "sadness", "anger", "fear", "surprise", "neutral", "love"]
        self.loaded = False
        self._load_model()

    def _load_model(self):
        """
        Load pre-trained transformer model
        In production: Use transformers library with distilbert-base-uncased-finetuned-sst-2-english
        """

        try:
            # Placeholder for actual model loading
            # from transformers import pipeline
            # self.sentiment_analyzer = pipeline("sentiment-analysis")
            # self.emotion_classifier = pipeline("text-classification", model="j-hartmann/emotion-english-distilroberta-base")

            print("[OK] Text emotion model initialized (mock mode)")
            self.loaded = True
        except Exception as e:
            print(f"[WARNING] Text model loading failed: {e}")
            self.loaded = False

    def predict_emotion(self, text: str) -> Dict:
        """

```

Predict emotion from text input

Args:

text: Input text string

Returns:

Dictionary with emotion, confidence, all_scores, and sentiment

"""

Clean text

text = self._preprocess_text(text)

Mock emotion detection (replace with actual model inference)

emotion_scores = self.mock_emotion_prediction(text)

Get dominant emotion

dominant_emotion = max(emotion_scores, key=emotion_scores.get)

confidence = emotion_scores[dominant_emotion]

Sentiment analysis

sentiment = self.analyze_sentiment(emotion_scores)

return {

"emotion": dominant_emotion,

"confidence": float(confidence),

"all_scores": {k: float(v) for k, v in emotion_scores.items()},

"sentiment": sentiment

}

def _preprocess_text(self, text: str) -> str:

"""Clean and normalize text"""

text = text.strip()

```
text = re.sub(r'\s+', ' ', text)
```

```
return text
```

```
def _mock_emotion_prediction(self, text: str) -> Dict[str, float]:
```

```
"""
```

```
Mock emotion prediction based on keyword matching
```

```
Replace with actual transformer model inference in production
```

```
"""
```

```
text_lower = text.lower()
```

```
scores = {emotion: 0.1 for emotion in self.emotions}
```

```
# Keyword-based heuristics for demo
```

```
if any(word in text_lower for word in ['happy', 'great', 'wonderful', 'amazing', 'excellent']):
```

```
scores['joy'] += 0.5
```

```
if any(word in text_lower for word in ['sad', 'depressed', 'down', 'unhappy', 'disappointed']):
```

```
scores['sadness'] += 0.5
```

```
if any(word in text_lower for word in ['angry', 'furious', 'mad', 'irritated', 'frustrated']):
```

```
scores['anger'] += 0.5
```

```
if any(word in text_lower for word in ['scared', 'afraid', 'worried', 'anxious', 'nervous']):
```

```
scores['fear'] += 0.5
```

```
if any(word in text_lower for word in ['wow', 'surprised', 'shocked', 'unexpected', 'amazing']):
```

```
scores['surprise'] += 0.3
```

```
if any(word in text_lower for word in ['love', 'adore', 'cherish', 'care', 'affection']):
```

```
scores['love'] += 0.5
```

```
# Normalize scores
```

```

total = sum(scores.values())
    normalized_scores = {k: v/total for k, v in scores.items()}

    return normalized_scores

def _analyze_sentiment(self, emotion_scores: Dict[str, float]) -> str:
    """Determine overall sentiment from emotion distribution"""
    positive_emotions = ['joy', 'love', 'surprise']
    negative_emotions = ['sadness', 'anger', 'fear']

    positive_score = sum (emotion_scores.get(e, 0) for e in positive_emotions)
    negative_score = sum (emotion_scores.get(e, 0) for e in negative_emotions)

    if positive_score > negative_score + 0.1:
        return "positive"
    elif negative_score > positive_score + 0.1:
        return "negative"
    else:
        return "neutral"

def extract_keywords(self, text: str, top_k: int = 5) -> List[str]:
    """Extract key emotional words from text"""
    # Simple keyword extraction (replace with TF-IDF or KeyBERT in production)
    emotional_words = [
        'happy', 'sad', 'angry', 'love', 'fear', 'joy', 'anxiety',
        'excited', 'depressed', 'calm', 'stressed', 'relaxed'
    ]

    words = text.lower().split()
    keywords = [word for word in words if word in emotional_words]

```

```
return keywords[:top_k]
```

```
def generate_insight(self, emotion: str, scores: Dict[str, float]) -> str:
```

```
    """Generate human-readable insight from emotion analysis"""
```

```
    insights = {
```

```
        "joy": "You're expressing positive emotions. Keep embracing this energy!",
```

```
        "sadness": "You seem to be going through a tough time. Remember, it's okay to feel this way.",
```

```
        "anger": "There's frustration in your words. Consider taking a moment to breathe.",
```

```
        "fear": "You're experiencing anxiety. Try grounding techniques or talking to someone.",
```

```
        "love": "Beautiful expressions of affection. Connection is powerful.",
```

```
        "neutral": "Your emotional state seems balanced right now.",
```

```
        "surprise": "Something unexpected caught your attention!"
```

```
    }
```

```
    return insights.get(emotion, "Your emotional state is being processed.")
```

```
    """
```

```
Voice Emotion Model
```

```
Speech Emotion Recognition using audio features and ML models
```

```
    """
```

```
import random
```

```
from typing import Dict
```

```
import io
```

```
class VoiceEmotionModel:
```

```
    """
```

```
    Voice-based emotion recognition from audio input
```

```
    Uses mel-spectrograms and audio features for emotion classification
```

```
    """
```

```
def __init__(self):
```

```

self.emotions = ["neutral", "happy", "sad", "angry", "fear", "disgust", "surprise"]
self.loaded = False
self._load_model()

```

```

def _load_model(self):

```

```

    """

```

```

    Load pre-trained voice emotion model

```

```

    In production: Use speechbrain/emotion-recognition or similar

```

```

    """

```

```

    try:

```

```

        # Placeholder for actual model loading

```

```

        # import torchaudio

```

```

        # from speechbrain.pretrained import EncoderClassifier

```

```

        # self.classifier = EncoderClassifier.from_hparams(

```

```

            # source="speechbrain/emotion-recognition-wav2vec2-IEMOCAP"

```

```

        # )

```

```

    print("✓ Voice emotion model initialized (mock mode)")

```

```

    self.loaded = True

```

```

    except Exception as e:

```

```

        print(f"⚠ Voice model loading failed: {e}")

```

```

    self.loaded = False

```

```

def predict_emotion(self, audio_bytes: bytes) -> Dict:

```

```

    """

```

```

    Predict emotion from audio input

```

```

    Args:

```

```

    audio_bytes: Raw audio file bytes

```

```

    Returns:

```

Dictionary with emotion, confidence, all_scores, and audio_features

"""

Extract audio features

features = self.extract_audio_features(audio_bytes)

Mock emotion prediction (replace with actual model inference)

emotion_scores = self.mock_voice_prediction(features)

Get dominant emotion

dominant_emotion = max(emotion_scores, key=emotion_scores.get)

confidence = emotion_scores[dominant_emotion]

return {

"emotion": dominant_emotion,

"confidence": float(confidence),

"all_scores": {k: float(v) for k, v in emotion_scores.items()},

"audio_features": features

}

def predict_emotion_fast(self, audio_bytes: bytes) -> Dict:

"""

Fast emotion prediction for real-time streaming

Uses simplified feature extraction

"""

features = self._extract_audio_features_fast(audio_bytes)

emotion_scores = self._mock_voice_prediction(features)

dominant_emotion = max(emotion_scores, key=emotion_scores.get)

return {

"emotion": dominant_emotion,

"confidence": float(emotion_scores[dominant_emotion])

```

}

def _extract_audio_features(self, audio_bytes: bytes) -> Dict:
    """
    Extract comprehensive audio features for emotion recognition

    Features include:
    - Pitch (fundamental frequency)
    - Energy/Intensity
    - Tempo/Speaking rate
    - Spectral features
    - MFCCs (Mel-frequency cepstral coefficients)
    """
    # In production, use librosa or torchaudio
    # import librosa
    # y, sr = librosa.load(io.BytesIO(audio_bytes))
    # mfccs = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=13)
    # pitch = librosa.yin(y, fmin=50, fmax=400)
    # energy = librosa.feature.rms(y=y)

    # Mock features for demo
    features = {
        "duration_seconds": random.uniform(1, 10),
        "pitch_mean": random.uniform(100, 250),
        "pitch_std": random.uniform(10, 50),
        "energy_mean": random.uniform(0.3, 0.9),
        "energy_std": random.uniform(0.05, 0.2),
        "speaking_rate": random.uniform(2, 5),
        "mfcc_mean": [random.uniform(-50, 50) for _ in range(13)],
        "spectral_centroid": random.uniform(1000, 3000)
    }
    return features

```



```

def _extract_audio_features_fast(self, audio_bytes: bytes) -> Dict:
    """Fast feature extraction for real-time processing"""
    return {
        "energy_mean": random.uniform(0.3, 0.9),
        "pitch_mean": random.uniform(100, 250)
    }

def _mock_voice_prediction(self, features: Dict) -> Dict[str, float]:
    """
    Mock voice emotion prediction
    Replace with actual model inference in production
    """

    scores = {emotion: 0.1 for emotion in self.emotions}
    # Simple heuristics based on features
    energy = features.get("energy_mean", 0.5)
    pitch = features.get("pitch_mean", 150)
    # High energy + high pitch = happy/excited
    if energy > 0.7 and pitch > 200:
        scores['happy'] += 0.5
    # Low energy + low pitch = sad
    elif energy < 0.4 and pitch < 150:
        scores['sad'] += 0.5

    # High energy + moderate-high pitch = angry
    elif energy > 0.7 and 150 <= pitch <= 220:
        scores['angry'] += 0.4
    return normalized_scores

```

APPENDIX B

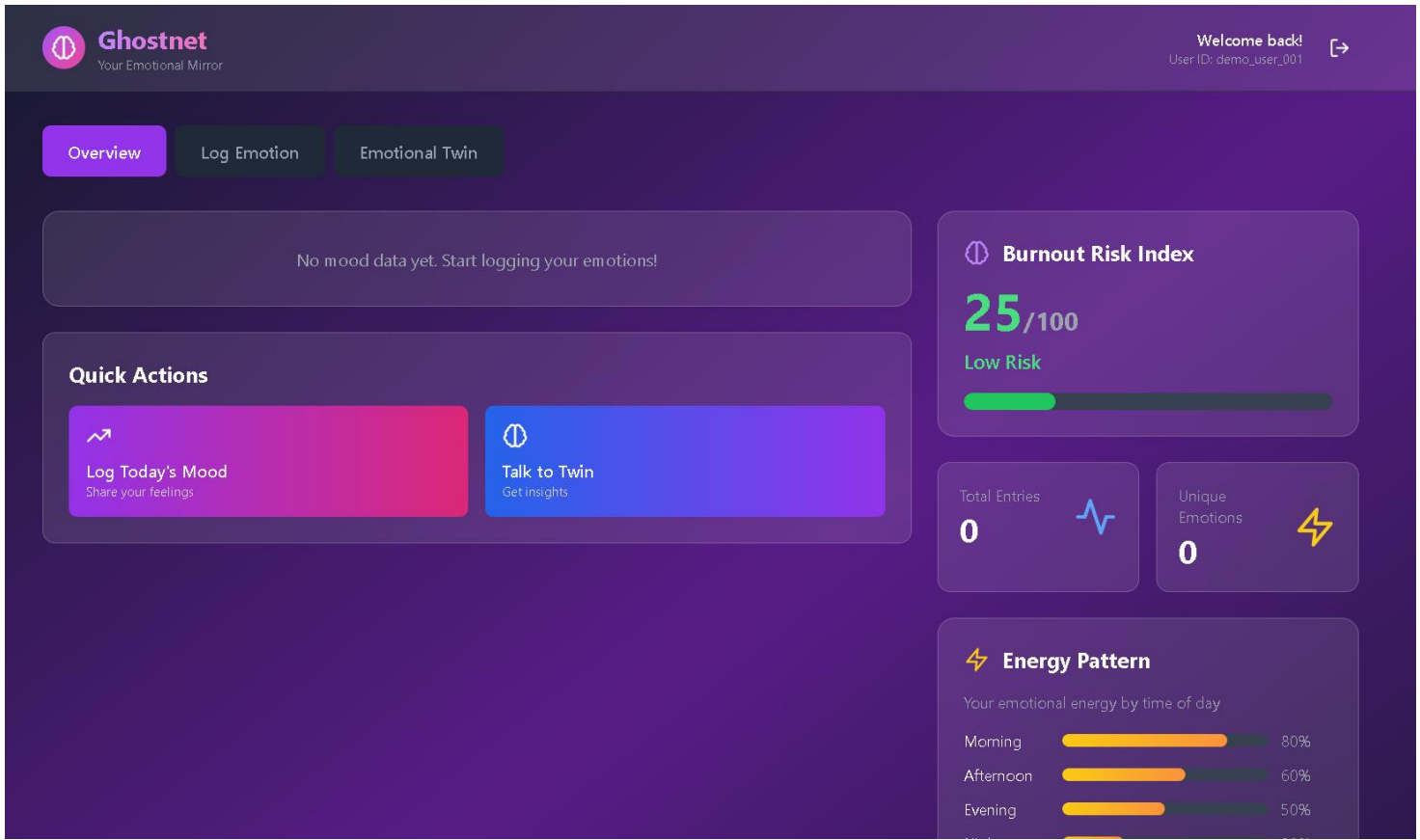


Fig 1.1: Dashboard Page

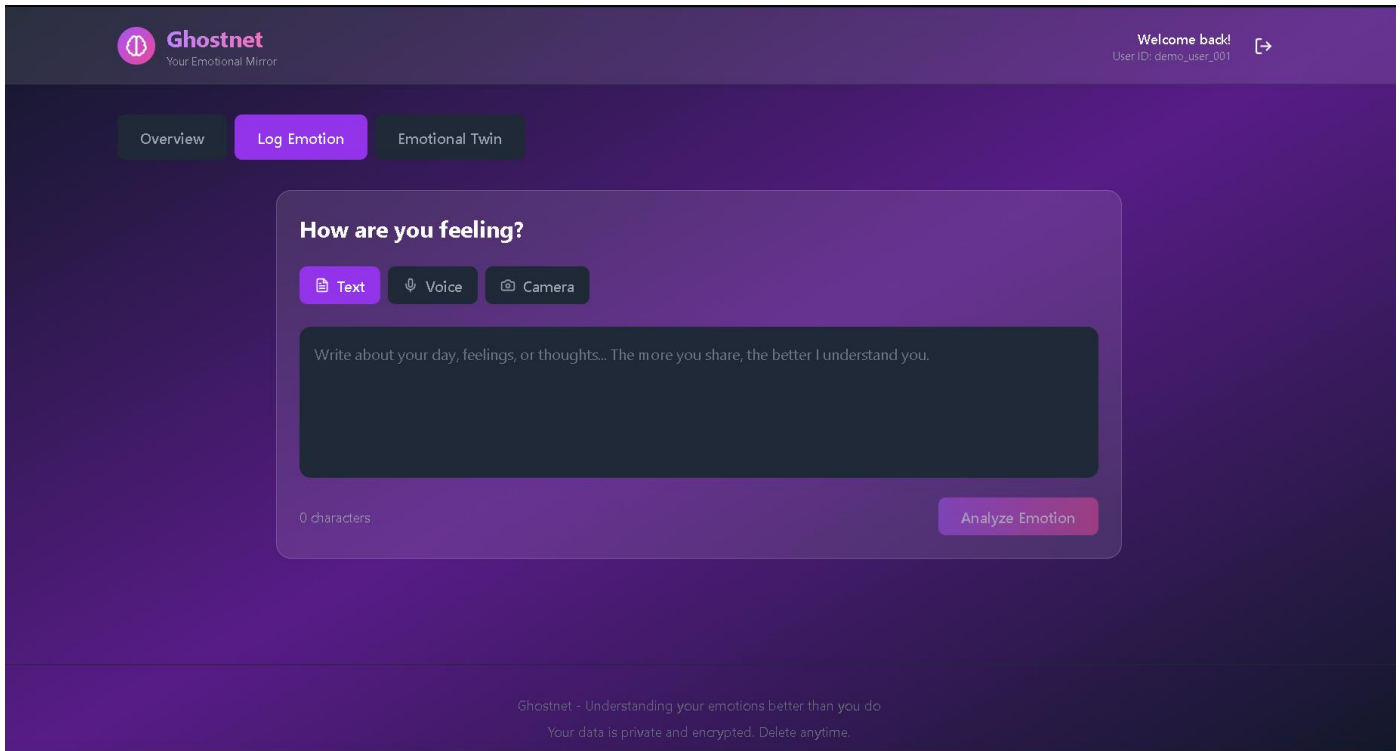


Fig 1.2:ContentPage

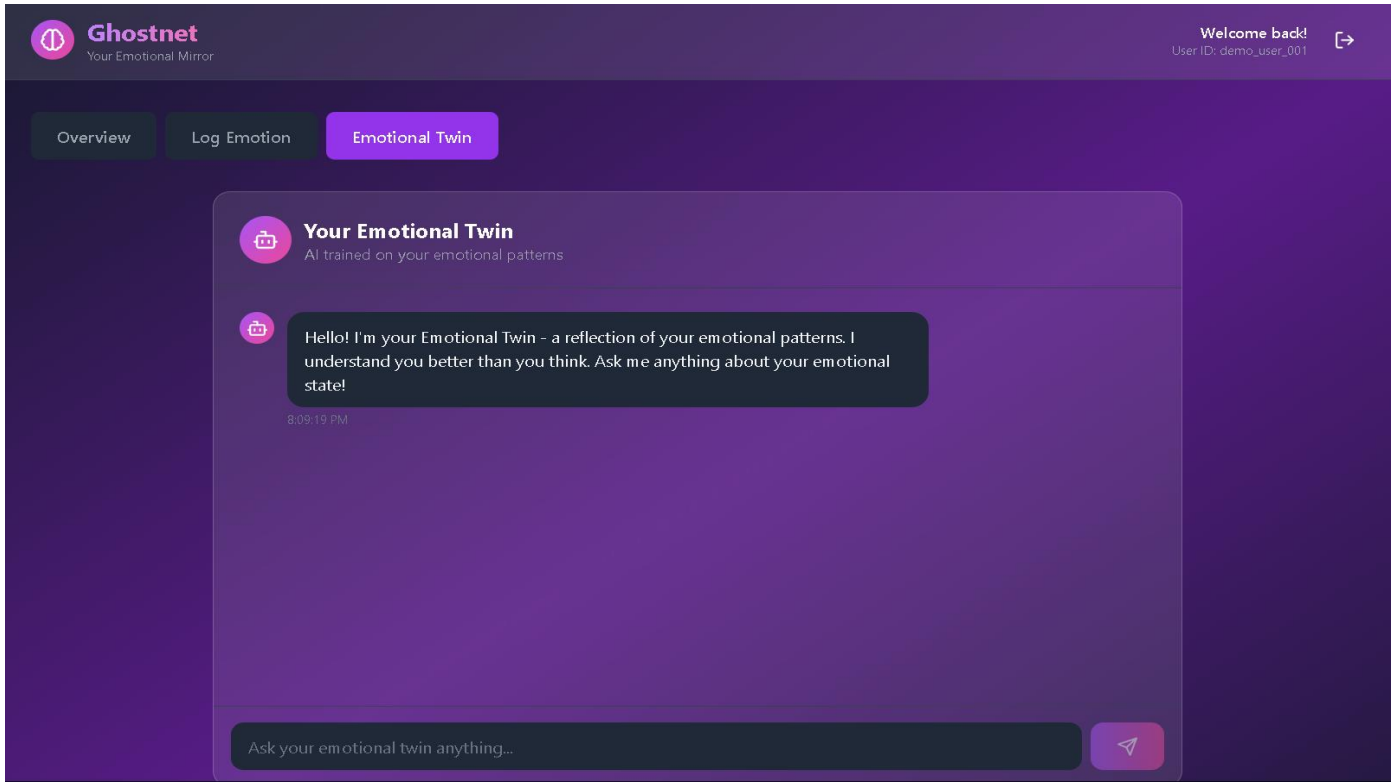


Fig 1.3: Chatbot Page

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LIST OF ABBREVIATION

NLP	Natural Language Processing
API	Application Program Interface
XML	Extensible Markup Language
AI	Artificial Intelligence
HCI	Human – Computer Interaction
CNN	Conceptual Neural Network
UI	User Interface
ML	Machine Learning
LSTM	Long Short-Term Memory
LLM	Large Language Model

